

Exercício 1

a)

$$p(x) = y_0 L_0 + y_1 L_1 + y_2 L_2 + y_3 L_3$$

$$x_1 = 1$$

$$x_2 = 10$$

$$x_3 = 100$$

$$x_4 = 1000$$

$$f(x_1) = \log_{10} x = f(1) = \log_{10}^1 = 0$$

$$f(x_2) = \log_{10} x = f(10) = \log_{10}^{10} = 1$$

$$f(x_3) = \log_{10} x = f(100) = \log_{10}^{100} = 2$$

$$f(x_4) = \log_{10} x = f(1000) = \log_{10}^{1000} = 3$$

b)

$$x_0 = 1$$

$$x_1 = 10$$

$$x_2 = 100$$

$$y_0 = 0$$

$$y_1 = 1$$

$$y_2 = 2$$

$$p(x) = y_0 L_0 + y_1 L_1 + y_2 L_2$$

$$L_0 = \frac{x-x_1}{x_0-x_1} \cdot \frac{x-x_2}{x_0-x_2} = \frac{x-10}{1-10} \cdot \frac{x-100}{1-100} = \frac{x-10}{-9} \cdot \frac{x-100}{-99}$$

$$L_1 = \frac{x-x_0}{x_1-x_0} \cdot \frac{x-x_2}{x_1-x_2} = \frac{x-1}{10-1} \cdot \frac{x-100}{10-100} = \frac{x-1}{9} \cdot \frac{x-100}{-90}$$

$$L_2 = \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1} = \frac{x-1}{100-1} \cdot \frac{x-10}{100-10} = \frac{x-1}{99} \cdot \frac{x-10}{90}$$

$$p(x) = \left(\frac{x-1}{9} \cdot \frac{x-100}{-99} \right) + 2 \left(\frac{x-1}{99} \cdot \frac{x-10}{90} \right)$$

c)

$$x_0 = 1$$

$$y_0 = 0$$

$$x_1 = 10$$

$$y_1 = 1$$

$$x_2 = 1000$$

$$y_2 = 3$$

$$L_0(x) = \frac{x-10}{1-10} \cdot \frac{x-1000}{1-1000} = \frac{x-10}{-9} \cdot \frac{x-1000}{-999}$$

$$L_1(x) = \frac{x-1}{10-1} \cdot \frac{x-1000}{10-1000} = \frac{x-1}{9} \cdot \frac{x-1000}{-990}$$

$$L_2(x) = \frac{x-1}{1000-1} \cdot \frac{x-10}{1000-10} = \frac{x-1}{999} \cdot \frac{x-10}{990}$$

$$p(x) = \frac{x-1}{9} \cdot \frac{x-1000}{-999} + 3 \left(\frac{x-1}{999} \cdot \frac{x-10}{990} \right)$$

$$d) f(500) = \log_{10}(500) = \underline{2,69897},$$

$$p_1(500) = \left(\frac{500-1}{9} \cdot \frac{500-100}{-90} \right) + 2 \left(\frac{500-1}{99} \cdot \frac{500-10}{90} \right)$$

$$\rightarrow \frac{499 \cdot 400}{9 \cdot -90} + 2 \left(\frac{499 \cdot 490}{99 \cdot 90} \right)$$

$$\rightarrow \frac{199.600}{-810} + 2 \left(\frac{244.510}{8910} \right)$$

$$\rightarrow -246,4 + 2 \cdot 27,4$$

$$\underline{p_1(500) = -191,5156},$$

$$p_2(500) = \frac{500-1}{9} \cdot \frac{500-1000}{-990} + 3 \left(\frac{500-1}{999} \cdot \frac{500-10}{990} \right)$$

$$\rightarrow \frac{499}{9} \cdot \frac{-500}{-990} + 3 \left(\frac{499}{999} \cdot \frac{490}{990} \right)$$

$$\frac{-249.500}{-8910} + 3 \left(\frac{244.510}{989.010} \right) = \frac{28,0}{28,0} + \frac{3 \cdot 0,247}{0,7416} = 28,74$$

$$\underline{p_2(500) = 28,74},$$

e) polinômio $p_2(x)$ aproxima-se mais de $f(500)$ do que $p_1(x)$.

$$f) p_1(1000) = \left(\frac{1000-1}{-810} \cdot \frac{1000-100}{-810} \right) + 2 \left(\frac{1000-1}{8910} \cdot \frac{1000-10}{8910} \right)$$

$$\Rightarrow \frac{-1.110}{-1110} + \frac{2 \cdot 111}{222}$$

$$\underline{p_1(1000) = -888}, \text{ Não modela bem.}$$

$$g) p_2(100) = \left(\frac{100-1}{-8910} \cdot \frac{100-1000}{-8910} \right) + 3 \left(\frac{100-1}{989.010} \cdot \frac{100-10}{989.010} \right) \text{ Não modela bem.}$$

$$-0,998 + 0,027 = -0,970 \quad \underline{p_2(100) = -0,970},$$